Image Processing Lab: From Pixels to Perception

ITAI 1378 Module 04 - Hands-On Laboratory

Duration: 45 minutes
Platform: Google Colab (Free Version)
Prerequisites: Basic Python knowledge from previous classes

🎯 Lab Objectives

By the end of this lab, you will:

1. **Understand** how computers represent images as numerical matrices
2. **Implement** fundamental image processing operations from scratch
3. **Apply** traditional image processing techniques using OpenCV and Pillow
4. **Experience** the connection between theory and practice
5. **Prepare** for your Image Processing Adventure Quest assignment

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📚 Connection to Today's Lecture

This lab directly implements the concepts we explored in class:

- **Digital Image Representation**: See how images become matrices of numbers
- **Core Operations**: Implement point, neighborhood, and geometric operations
- **Traditional Tools**: Hands-on experience with OpenCV and Pillow
- **Bridge to AI**: Understand the foundation that powers modern AI tools like Nano Banana

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⏱️ Lab Timeline (45 minutes)

Time	Section	Activity
0-5 min	Setup	Environment setup and imports
5-15 min	Part 1	Digital image fundamentals
15-25 min	Part 2	Basic image operations
25-35 min	Part 3	Advanced processing techniques
35-40 min	Part 4	Creative exploration
40-45 min	Wrap-up	Reflection and next steps

🚀 Let's Begin!

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Important: Run each cell in order. Read the explanations carefully - they connect directly to today's lecture concepts!

🔧 Setup and Environment Preparation

First, let's set up our environment. We'll use the same libraries we discussed in class:

- **OpenCV**: The powerhouse for computer vision (remember from our tools discussion?)
- **Pillow (PIL)**: Python-friendly image processing
- **NumPy**: For matrix operations (images are matrices!)
- **Matplotlib**: For displaying our results

💡 **Recall from lecture:** These are the "traditional tools" that give us precise control over every pixel!

```
# Install required packages (Google Colab already has most of these)
!pip install opencv-python-headless pillow matplotlib numpy

# Import all necessary libraries
import cv2                    # OpenCV for computer vision
import numpy as np           # NumPy for numerical operations on matrices
from PIL import Image, ImageFilter, ImageEnhance  # Pillow for image process
import matplotlib.pyplot as plt  # For displaying images
import matplotlib.patches as patches  # For drawing on images
from google.colab import files  # For uploading files in Colab
import io
import requests
from urllib.parse import urlparse

# Configure matplotlib for better image display
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['image.cmap'] = 'gray'  # Default to grayscale colormap

print("✅ Environment setup complete!")
print(f"📦 OpenCV version: {cv2.__version__}")
print(f"📦 NumPy version: {np.__version__}")
print("🎉 Ready to explore image processing!")
```

```
Requirement already satisfied: opencv-python-headless in /usr/local/lib/pytho
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/
```

```
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pac
✅ Environment setup complete!
📦 OpenCV version: 4.13.0
📦 NumPy version: 2.0.2
🎉 Ready to explore image processing!
```

📷 Load Sample Images

Let's load some sample images to work with. We'll provide both options:

1. **Download sample images** from the internet
2. **Upload your own images** (optional)

🔗 **Connection to lecture:** Remember how we discussed image acquisition as the first step in any vision system?

```
def download_sample_image(url, filename):
    """
    Download a sample image from URL for our experiments.
    This simulates the 'image acquisition' step we discussed in class.
    """
    try:
        response = requests.get(url)
        if response.status_code == 200:
            with open(filename, 'wb') as f:
                f.write(response.content)
            print(f"✅ Downloaded: {filename}")
            return True
    except Exception as e:
        print(f"❌ Error downloading {filename}: {e}")
    return False

# Download sample images for our experiments
sample_images = {
    'landscape.jpg': 'https://upload.wikimedia.org/wikipedia/commons/thumb/5
    'portrait.jpg': 'https://upload.wikimedia.org/wikipedia/commons/thumb/5/
}

# Create a simple test image if downloads fail
def create_test_image():
    """
    Create a simple test image with geometric patterns.
    This demonstrates how images are just matrices of numbers!
```

```
    This demonstrates how images are just matrices of numbers.
    """
    # Create a 200x200 image with interesting patterns
    img = np.zeros((200, 200, 3), dtype=np.uint8)

    # Add some geometric shapes (remember: we're just setting pixel values!)
    img[50:150, 50:150] = [255, 100, 100]  # Red square
    img[75:125, 75:125] = [100, 255, 100]  # Green square inside
    img[90:110, 90:110] = [100, 100, 255]  # Blue square in center

    return img

# Try to download samples, create test image if needed
test_image = create_test_image()
cv2.imwrite('test_image.jpg', cv2.cvtColor(test_image, cv2.COLOR_RGB2BGR))

print("🖼️ Sample images ready for processing!")
print("📝 Note: We created a test image to ensure you have something to worl
```

```
🖼️ Sample images ready for processing!
📝 Note: We created a test image to ensure you have something to work with.
```

📊 Part 1: Digital Image Fundamentals (10 minutes)

Understanding Images as Matrices

🎓 **Lecture Connection:** Remember when we discussed how computers see images as matrices of numbers? Let's see this in action!

In our lecture, we learned:

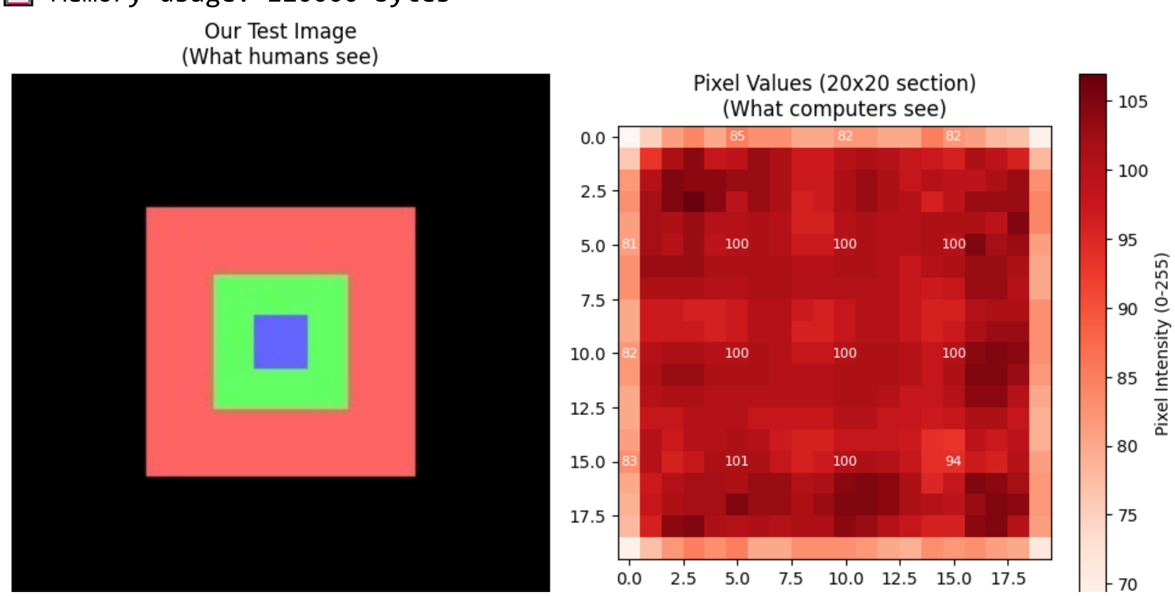
- **Grayscale images**: 2D matrices (height × width)
- **Color images**: 3D matrices (height × width × channels)
- **Pixel values**: Numbers from 0-255 for 8-bit images

Let's explore this hands-on!

```
# Load our test image using OpenCV
# Note: OpenCV loads images in BGR format (Blue, Green, Red)
img_bgr = cv2.imread('test_image.jpg')

# Convert to RGB for proper display (remember color spaces from lecture?)
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

# Let's examine the image properties
print("🔍 IMAGE ANALYSIS - Just like we discussed in class!")
print(f"📐 Image shape: {img_rgb.shape}")
print(f"📊 Data type: {img_rgb.dtype}")
```

```
print(f"📊 Data type: {img_rgb.dtype}")
print(f"📈 Value range: {img_rgb.min()} to {img_rgb.max()}")
print(f"💾 Memory usage: {img_rgb.nbytes} bytes")

# Display the image
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title('Our Test Image\n(What humans see)')
plt.axis('off')

plt.subplot(1, 2, 2)
# Show a small section of pixel values (what computers see)
pixel_section = img_rgb[90:110, 90:110, 0]  # 20x20 red channel section
plt.imshow(pixel_section, cmap='Reds')
plt.title('Pixel Values (20x20 section)\n(What computers see)')
plt.colorbar(label='Pixel Intensity (0-255)')

# Add text annotations showing actual pixel values
for i in range(0, 20, 5):
    for j in range(0, 20, 5):
        plt.text(j, i, str(pixel_section[i, j]),
                ha='center', va='center', color='white', fontsize=8)

plt.tight_layout()
plt.show()

print("\n💡 Key Insight: The image on the left is what we see.")
print("   The numbers on the right are what the computer sees!")
```

```
🔍 IMAGE ANALYSIS - Just like we discussed in class!
📐 Image shape: (200, 200, 3)
📊 Data type: uint8
📈 Value range: 0 to 255
💾 Memory usage: 120000 bytes
```



```
💡 Key Insight: The image on the left is what we see.
   The numbers on the right are what the computer sees!
```

Exploring RGB Channels

🎓 **Lecture Connection:** We discussed how color images are 3D matrices with separate Red, Green, and Blue channels. Let's separate them and see each channel individually!

```python
# Separate the RGB channels (remember the 3D matrix concept?)
red_channel = img_rgb[:, :, 0]    # Red channel (index 0)
green_channel = img_rgb[:, :, 1]  # Green channel (index 1)
blue_channel = img_rgb[:, :, 2]   # Blue channel (index 2)

# Create visualizations of each channel
plt.figure(figsize=(15, 10))

# Original image
plt.subplot(2, 3, 1)
plt.imshow(img_rgb)
plt.title('Original Image\n(All channels combined)')
plt.axis('off')

# Red channel
plt.subplot(2, 3, 2)
plt.imshow(red_channel, cmap='Reds')
plt.title('Red Channel\n(Higher values = more red)')
plt.axis('off')
plt.colorbar(shrink=0.8)

# Green channel
plt.subplot(2, 3, 3)
plt.imshow(green_channel, cmap='Greens')
plt.title('Green Channel\n(Higher values = more green)')
plt.axis('off')
plt.colorbar(shrink=0.8)

# Blue channel
plt.subplot(2, 3, 4)
plt.imshow(blue_channel, cmap='Blues')
plt.title('Blue Channel\n(Higher values = more blue)')
plt.axis('off')
plt.colorbar(shrink=0.8)
```

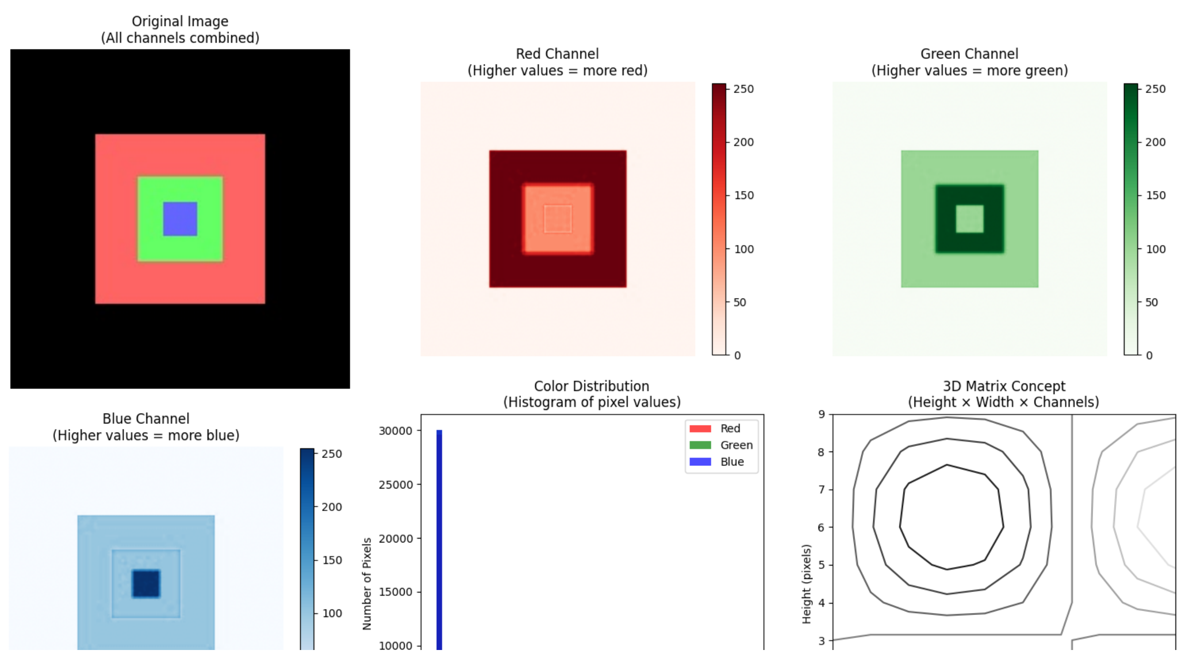
```
# Histogram of all channels
plt.subplot(2, 3, 5)
plt.hist(red_channel.flatten(), bins=50, alpha=0.7, color='red', label='Red'
plt.hist(green_channel.flatten(), bins=50, alpha=0.7, color='green', label='
plt.hist(blue_channel.flatten(), bins=50, alpha=0.7, color='blue', label='Bl
plt.title('Color Distribution\n(Histogram of pixel values)')
plt.xlabel('Pixel Intensity (0-255)')
plt.ylabel('Number of Pixels')
plt.legend()

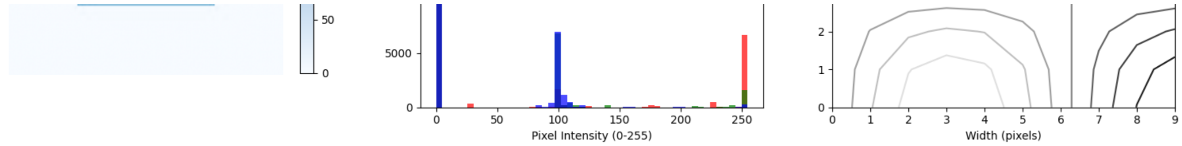
# 3D representation concept
plt.subplot(2, 3, 6)
# Create a simple 3D visualization concept
x = np.arange(0, 10)
y = np.arange(0, 10)
X, Y = np.meshgrid(x, y)
Z = np.sin(X/2) * np.cos(Y/2)
plt.contour(X, Y, Z)
plt.title('3D Matrix Concept\n(Height × Width × Channels)')
plt.xlabel('Width (pixels)')
plt.ylabel('Height (pixels)')

plt.tight_layout()
plt.show()

# Print some statistics
print("📊 CHANNEL STATISTICS:")
print(f"🔴 Red channel   - Min: {red_channel.min():3d}, Max: {red_channel.ma
print(f"🟢 Green channel - Min: {green_channel.min():3d}, Max: {green_channe
print(f"🔵 Blue channel  - Min: {blue_channel.min():3d}, Max: {blue_channel.

print("\n💡 Notice how different parts of the image contribute different amo
```






```
📊 CHANNEL STATISTICS:
🔴 Red channel   - Min:   0, Max: 255, Mean: 53.9
🟢 Green channel - Min:   0, Max: 255, Mean: 33.3
🔵 Blue channel  - Min:   0, Max: 255, Mean: 26.7

💡 Notice how different parts of the image contribute different amounts to ea
```

Converting to Grayscale

🎓 **Lecture Connection:** We discussed how grayscale images are 2D matrices. Let's convert our color image to grayscale and see the difference!

```
# Convert to grayscale using different methods
# Method 1: OpenCV conversion (uses weighted average)
gray_opencv = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2GRAY)

# Method 2: Simple average (R + G + B) / 3
gray_average = np.mean(img_rgb, axis=2).astype(np.uint8)

# Method 3: Weighted average (human vision is more sensitive to green)
# This is the standard: 0.299*R + 0.587*G + 0.114*B
gray_weighted = (0.299 * img_rgb[:,:,0] +
                 0.587 * img_rgb[:,:,1] +
                 0.114 * img_rgb[:,:,2]).astype(np.uint8)

# Display the results
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(img_rgb)
plt.title('Original Color Image\n(3D matrix: H×W×3)')
plt.axis('off')

plt.subplot(1, 4, 2)
plt.imshow(gray_opencv, cmap='gray')
plt.title('OpenCV Grayscale\n(2D matrix: H×W)')
plt.axis('off')

plt.subplot(1, 4, 3)
plt.imshow(gray_average, cmap='gray')
```

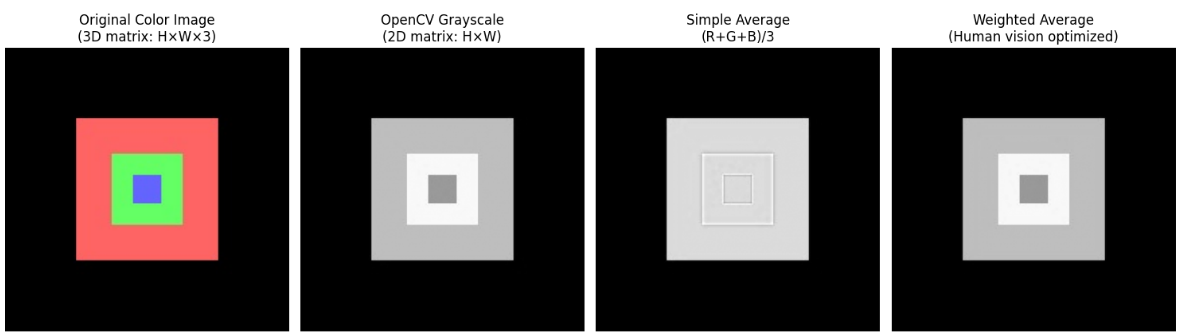
```
    plt.title('Simple Average\n(R+G+B)/3')
    plt.axis('off')

    plt.subplot(1, 4, 4)
    plt.imshow(gray_weighted, cmap='gray')
    plt.title('Weighted Average\n(Human vision optimized)')
    plt.axis('off')

    plt.tight_layout()
    plt.show()

    # Compare the shapes
    print("📐 SHAPE COMPARISON:")
    print(f"🎨 Original color image: {img_rgb.shape} (3D matrix)")
    print(f"⚫ Grayscale image:     {gray_opencv.shape} (2D matrix)")
    print(f"💾 Memory reduction:    {img_rgb.nbytes} → {gray_opencv.nbytes} byte

    print("\n💡 Key Insight: Grayscale conversion reduces data by 2/3 while pres
```




```
📐 SHAPE COMPARISON:
🎨 Original color image: (200, 200, 3) (3D matrix)
⚫ Grayscale image:     (200, 200) (2D matrix)
💾 Memory reduction:    120000 → 40000 bytes (33.3%)

💡 Key Insight: Grayscale conversion reduces data by 2/3 while preserving str
```

🔧 Part 2: Basic Image Operations (10 minutes)

Point Operations

🎓 **Lecture Connection:** Remember our discussion of point operations? These work on individual pixels without considering neighbors. Let's implement brightness and contrast adjustments!

```
    # Use our grayscale image for point operations
    img_gray = gray_opencv.copy()
```

```
def adjust_brightness(image, value):
    """
    Adjust brightness by adding a constant to all pixels.
    This is a classic 'point operation' - each pixel is processed independen

    Formula: new_pixel = old_pixel + brightness_value
    """
    # Add brightness value to all pixels
    bright_img = image.astype(np.int16) + value  # Use int16 to prevent over

    # Clip values to valid range [0, 255]
    bright_img = np.clip(bright_img, 0, 255)

    return bright_img.astype(np.uint8)

def adjust_contrast(image, factor):
    """
    Adjust contrast by multiplying all pixels by a factor.
    Another point operation - each pixel processed independently.

    Formula: new_pixel = old_pixel * contrast_factor
    """
    # Multiply all pixels by contrast factor
    contrast_img = image.astype(np.float32) * factor

    # Clip values to valid range [0, 255]
    contrast_img = np.clip(contrast_img, 0, 255)

    return contrast_img.astype(np.uint8)

# Apply different brightness and contrast adjustments
bright_up = adjust_brightness(img_gray, 50)      # Brighter
bright_down = adjust_brightness(img_gray, -50)   # Darker
contrast_up = adjust_contrast(img_gray, 1.5)     # Higher contrast
contrast_down = adjust_contrast(img_gray, 0.5)   # Lower contrast

# Display results
plt.figure(figsize=(15, 10))

images = [img_gray, bright_up, bright_down, contrast_up, contrast_down]
titles = ['Original', 'Brighter (+50)', 'Darker (-50)', 'High Contrast (×1.5

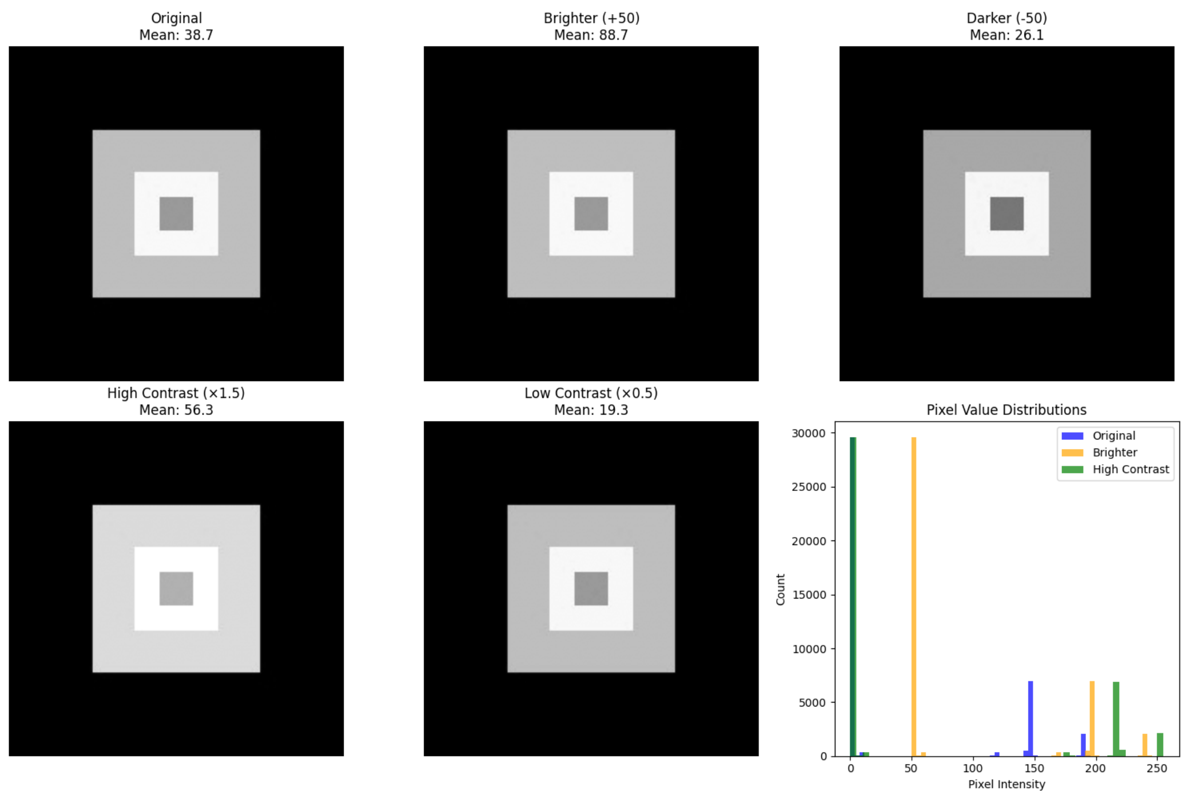
for i, (img, title) in enumerate(zip(images, titles)):
    plt.subplot(2, 3, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(f'{title}\nMean: {img.mean():.1f}')
    plt.axis('off')

# Show histograms to understand the changes
plt.subplot(2, 3, 6)
```

```
    plt.hist(img_gray.flatten(), bins=50, alpha=0.7, label='Original', color='bl
    plt.hist(bright_up.flatten(), bins=50, alpha=0.7, label='Brighter', color='o
    plt.hist(contrast_up.flatten(), bins=50, alpha=0.7, label='High Contrast', c
    plt.title('Pixel Value Distributions')
    plt.xlabel('Pixel Intensity')
    plt.ylabel('Count')
    plt.legend()

    plt.tight_layout()
    plt.show()

    print("💡 Point Operations Summary:")
    print("   • Brightness: Shifts the entire histogram left or right")
    print("   • Contrast: Stretches or compresses the histogram")
    print("   • Each pixel is processed independently (no neighbors considered)"
```




```
💡 Point Operations Summary:
   • Brightness: Shifts the entire histogram left or right
   • Contrast: Stretches or compresses the histogram
   • Each pixel is processed independently (no neighbors considered)
```

Neighborhood Operations (Filtering)

🎓 **Lecture Connection:** Now let's explore neighborhood operations! These consider a pixel's surrounding context. Convolution is the key operation here - remember our discussion about kernels and filters?

```
# Define different kernels (filters) for convolution
# Each kernel produces a different effect!

# Blur kernel (Gaussian-like)
blur_kernel = np.array([[1, 2, 1],
                        [2, 4, 2],
                        [1, 2, 1]]) / 16  # Normalize so sum = 1

# Edge detection kernel (Sobel horizontal)
edge_kernel_h = np.array([[-1, 0, 1],
                          [-2, 0, 2],
                          [-1, 0, 1]])

# Edge detection kernel (Sobel vertical)
edge_kernel_v = np.array([[-1, -2, -1],
                          [ 0,  0,  0],
                          [ 1,  2,  1]])

# Sharpening kernel
sharpen_kernel = np.array([[ 0, -1,  0],
                           [-1,  5, -1],
                           [ 0, -1,  0]])

def apply_kernel(image, kernel):
    """
    Apply a convolution kernel to an image.
    This is the fundamental 'neighborhood operation' from our lecture!
    """
    # Use OpenCV's filter2D function for convolution
    result = cv2.filter2D(image, -1, kernel)
    return result

# Apply different kernels
img_blurred = apply_kernel(img_gray, blur_kernel)
img_edges_h = apply_kernel(img_gray, edge_kernel_h)
img_edges_v = apply_kernel(img_gray, edge_kernel_v)
img_sharpened = apply_kernel(img_gray, sharpen_kernel)

# Combine horizontal and vertical edges
img_edges_combined = np.sqrt(img_edges_h.astype(np.float32)**2 +
                             img_edges_v.astype(np.float32)**2)
img_edges_combined = np.clip(img_edges_combined, 0, 255).astype(np.uint8)

# Display results
plt.figure(figsize=(15, 12))

# Original and processed images
```

```
images = [img_gray, img_blurred, img_edges_h, img_edges_v, img_edges_combine
titles = ['Original', 'Blurred\n(Gaussian-like)', 'Horizontal Edges\n(Sobel)
          'Vertical Edges\n(Sobel)', 'Combined Edges', 'Sharpened']

for i, (img, title) in enumerate(zip(images, titles)):
    plt.subplot(3, 3, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(title)
    plt.axis('off')

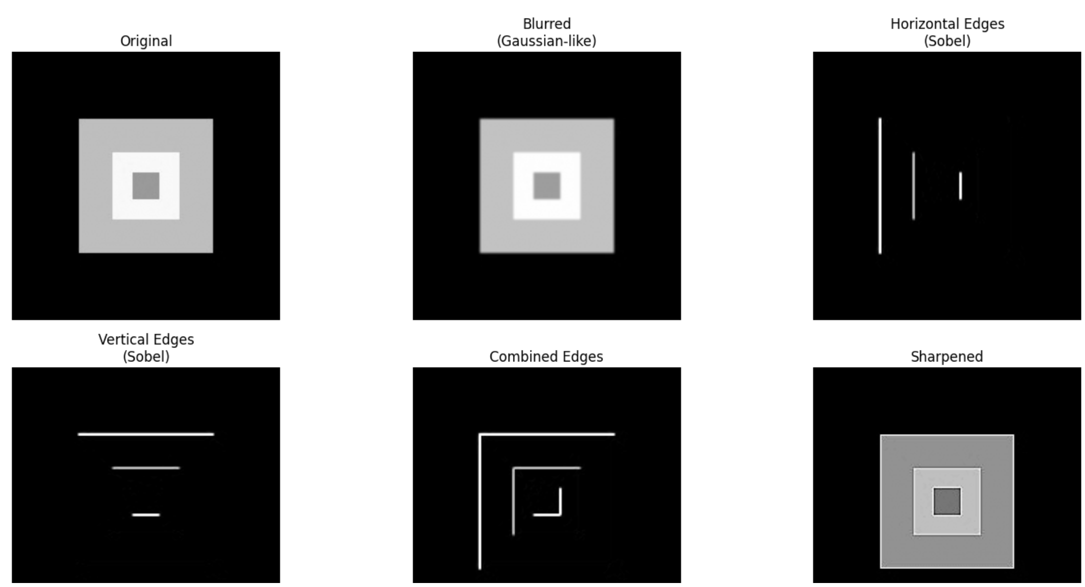
# Show the kernels
kernels = [blur_kernel, edge_kernel_h, sharpen_kernel]
kernel_titles = ['Blur Kernel', 'Edge Kernel (H)', 'Sharpen Kernel']

for i, (kernel, title) in enumerate(zip(kernels, kernel_titles)):
    plt.subplot(3, 3, 7+i)
    plt.imshow(kernel, cmap='RdBu', vmin=-2, vmax=2)
    plt.title(title)
    plt.colorbar(shrink=0.8)

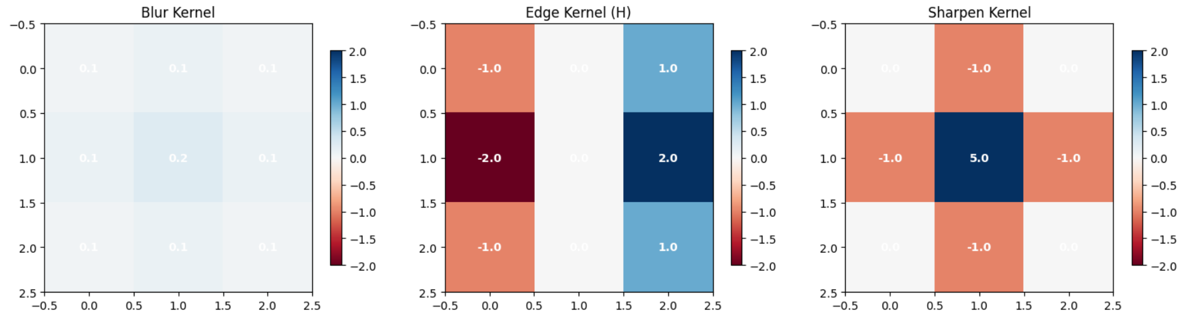
    # Add text annotations for kernel values
    for row in range(kernel.shape[0]):
        for col in range(kernel.shape[1]):
            plt.text(col, row, f'{kernel[row, col]:.1f}',
                    ha='center', va='center', color='white', fontweight='bol

plt.tight_layout()
plt.show()

print("🔍 Neighborhood Operations Summary:")
print("   • Blur: Averages neighboring pixels (smoothing)")
print("   • Edge Detection: Finds rapid changes in intensity")
print("   • Sharpening: Enhances differences between neighboring pixels")
print("   • Each operation uses a different kernel (filter matrix)")
print("\n💡 This is the foundation of how modern AI tools like Nano Banana w
```



[redacted] [redacted] [redacted]



🔍 Neighborhood Operations Summary:
- Blur: Averages neighboring pixels (smoothing)
- Edge Detection: Finds rapid changes in intensity
- Sharpening: Enhances differences between neighboring pixels
- Each operation uses a different kernel (filter matrix)

💡 This is the foundation of how modern AI tools like Nano Banana work!

🎨 Part 3: Advanced Processing Techniques (10 minutes)

Histogram Analysis and Enhancement

🎓 **Lecture Connection:** Remember our discussion of global operations? Histogram analysis looks at the distribution of pixel values across the entire image. Let's implement histogram equalization!

```
# Create a low-contrast image for demonstration
low_contrast_img = adjust_contrast(img_gray, 0.3)  # Very low contrast
low_contrast_img = adjust_brightness(low_contrast_img, 50)  # Shift to mid-r

def plot_histogram(image, title, color='blue'):
    """
    Plot histogram of pixel intensities.
    This shows the distribution of pixel values in the image.
    """
    hist, bins = np.histogram(image.flatten(), bins=256, range=[0, 256])
    plt.plot(bins[:-1], hist, color=color, alpha=0.7, linewidth=2)
    plt.title(title)
    plt.xlabel('Pixel Intensity (0-255)')
    plt.ylabel('Number of Pixels')
```

```
    plt.grid(True, alpha=0.3)

# Apply histogram equalization
equalized_img = cv2.equalizeHist(low_contrast_img)

# Apply CLAHE (Contrast Limited Adaptive Histogram Equalization)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
clahe_img = clahe.apply(low_contrast_img)

# Display results
plt.figure(figsize=(15, 12))

# Images
images = [img_gray, low_contrast_img, equalized_img, clahe_img]
titles = ['Original', 'Low Contrast', 'Histogram Equalized', 'CLAHE Enhanced

for i, (img, title) in enumerate(zip(images, titles)):
    # Show image
    plt.subplot(3, 4, i+1)
    plt.imshow(img, cmap='gray')
    plt.title(f'{title}\nMean: {img.mean():.1f}, Std: {img.std():.1f}')
    plt.axis('off')

    # Show histogram
    plt.subplot(3, 4, i+5)
    plot_histogram(img, f'{title} Histogram')
    plt.ylim(0, max(np.histogram(img_gray.flatten(), bins=256)[0]) * 1.1)

# Show cumulative distribution functions
plt.subplot(3, 4, 9)
for img, title, color in zip(images, titles, ['blue', 'red', 'green', 'orang
    hist, bins = np.histogram(img.flatten(), bins=256, range=[0, 256])
    cdf = hist.cumsum()
    cdf_normalized = cdf * hist.max() / cdf.max()  # Normalize for display
    plt.plot(cdf_normalized, color=color, label=title, alpha=0.7)
plt.title('Cumulative Distribution Functions')
plt.xlabel('Pixel Intensity')
plt.ylabel('Cumulative Count (normalized)')
plt.legend()
plt.grid(True, alpha=0.3)

# Show enhancement comparison
plt.subplot(3, 4, 10)
difference = equalized_img.astype(np.int16) - low_contrast_img.astype(np.int
plt.imshow(difference, cmap='RdBu', vmin=-100, vmax=100)
plt.title('Enhancement Difference\n(Equalized - Original)')
plt.colorbar(shrink=0.8)
plt.axis('off')

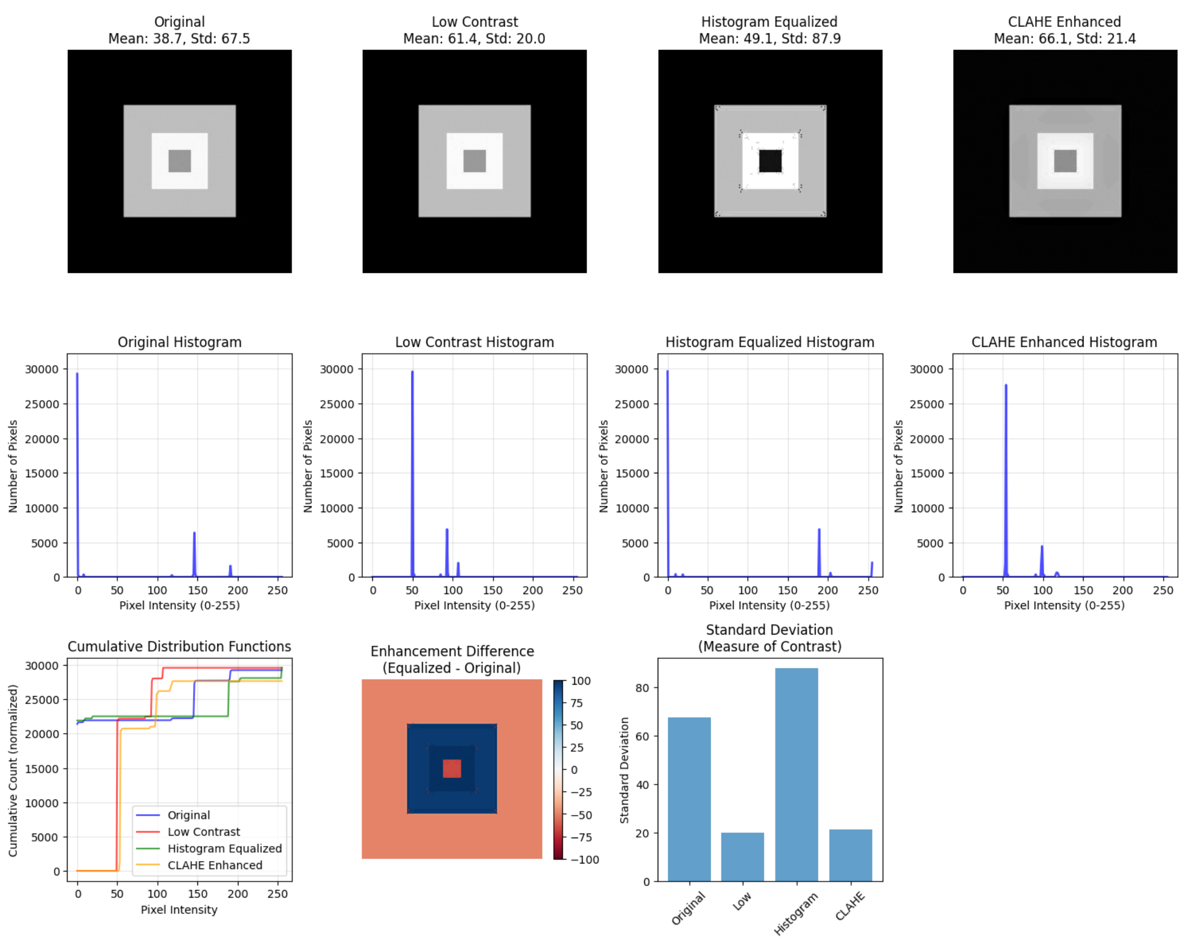
# Show statistics
plt.subplot(3, 4, 11)
```

```
stats_data = {
    'Image': titles,
    'Mean': [img.mean() for img in images],
    'Std': [img.std() for img in images],
    'Min': [img.min() for img in images],
    'Max': [img.max() for img in images]
}

x_pos = np.arange(len(titles))
plt.bar(x_pos, [img.std() for img in images], alpha=0.7)
plt.title('Standard Deviation\n(Measure of Contrast)')
plt.xlabel('Image Type')
plt.ylabel('Standard Deviation')
plt.xticks(x_pos, [t.split()[0] for t in titles], rotation=45)

plt.tight_layout()
plt.show()

print("📊 Histogram Enhancement Summary:")
print("   • Low contrast images have narrow histograms (pixels clustered in
print("   • Histogram equalization spreads pixels across full intensity rang
print("   • CLAHE prevents over-enhancement by limiting contrast in local re
print("   • Higher standard deviation = higher contrast")
print("\n💡 This is how your smartphone automatically improves photo contras
```



📊 Histogram Enhancement Summary:
- Low contrast images have narrow histograms (pixels clustered in small ra
- Histogram equalization spreads pixels across full intensity range
- CLAHE prevents over-enhancement by limiting contrast in local regions
- Higher standard deviation = higher contrast

💡 This is how your smartphone automatically improves photo contrast!

Geometric Transformations

🎓 **Lecture Connection:** Let's implement geometric operations that alter spatial relationships between pixels - scaling, rotation, and perspective correction!

```
# Use our original color image for geometric transformations
img_for_transform = img_rgb.copy()
height, width = img_for_transform.shape[:2]

# 1. Scaling (resize)
scale_factor = 0.7
new_width = int(width * scale_factor)
new_height = int(height * scale_factor)
scaled_img = cv2.resize(img_for_transform, (new_width, new_height), interpol

# 2. Rotation
rotation_angle = 30  # degrees
center = (width // 2, height // 2)
rotation_matrix = cv2.getRotationMatrix2D(center, rotation_angle, 1.0)
rotated_img = cv2.warpAffine(img_for_transform, rotation_matrix, (width, hei

# 3. Translation (shifting)
tx, ty = 30, 20  # shift by 30 pixels right, 20 pixels down
translation_matrix = np.float32([[1, 0, tx], [0, 1, ty]])
translated_img = cv2.warpAffine(img_for_transform, translation_matrix, (widt

# 4. Perspective transformation (simulating camera angle)
# Define source points (corners of original image)
src_points = np.float32([[0, 0], [width, 0], [width, height], [0, height]])
# Define destination points (perspective effect)
dst_points = np.float32([[20, 30], [width-10, 20], [width-30, height-20], [1
perspective_matrix = cv2.getPerspectiveTransform(src_points, dst_points)
perspective_img = cv2.warpPerspective(img_for_transform, perspective_matrix,

# 5. Shearing (skewing)
shear_factor = 0.2
shear_matrix = np.float32([[1, shear_factor, 0], [0, 1, 0]])
```

```
sheared_img = cv2.warpAffine(img_for_transform, shear_matrix, (width, height

# Display all transformations
plt.figure(figsize=(15, 12))

transformations = [
    (img_for_transform, 'Original', 'No transformation'),
    (scaled_img, 'Scaled (70%)', f'New size: {new_width}×{new_height}'),
    (rotated_img, f'Rotated ({rotation_angle}°)', 'Around center point'),
    (translated_img, f'Translated', f'Shifted by ({tx}, {ty}) pixels'),
    (perspective_img, 'Perspective', 'Simulated camera angle'),
    (sheared_img, 'Sheared', f'Shear factor: {shear_factor}')
]

for i, (img, title, subtitle) in enumerate(transformations):
    plt.subplot(2, 3, i+1)
    if img.shape[:2] != img_for_transform.shape[:2]:  # Handle different siz
        # Pad smaller images for consistent display
        pad_h = max(0, height - img.shape[0])
        pad_w = max(0, width - img.shape[1])
        img_padded = np.pad(img, ((0, pad_h), (0, pad_w), (0, 0)), mode='con
        plt.imshow(img_padded)
    else:
        plt.imshow(img)

    plt.title(f'{title}\n{subtitle}')
    plt.axis('off')

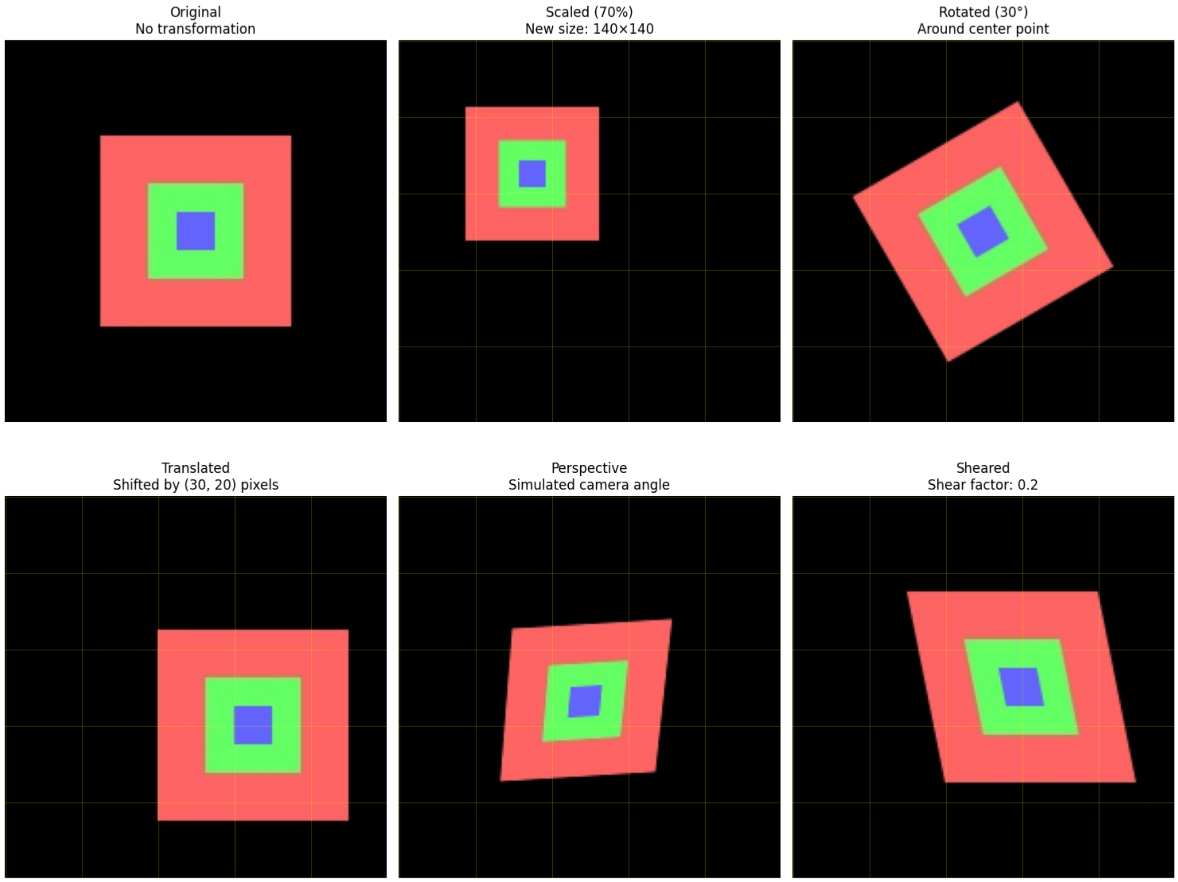
    # Add grid overlay to show transformation effect
    if i > 0:  # Skip original
        ax = plt.gca()
        # Add some reference lines
        for y in range(0, height, 40):
            ax.axhline(y=y, color='yellow', alpha=0.3, linewidth=0.5)
        for x in range(0, width, 40):
            ax.axvline(x=x, color='yellow', alpha=0.3, linewidth=0.5)

plt.tight_layout()
plt.show()

# Show transformation matrices
print("🔧 TRANSFORMATION MATRICES:")
print("\n📐 Rotation Matrix (30°):")
print(rotation_matrix)
print("\n➡️ Translation Matrix:")
print(translation_matrix)
print("\n🔄 Shear Matrix:")
print(shear_matrix)

print("\n💡 Geometric Transformations Summary:")
```

```
print("  • Scaling: Changes image size (interpolation needed)")
print("  • Rotation: Rotates around a center point")
print("  • Translation: Shifts image position")
print("  • Perspective: Simulates 3D viewing angle")
print("  • Shearing: Skews the image shape")
print("\n🎯 These are essential for image registration and augmentation!")
```




```
🔧 TRANSFORMATION MATRICES:

📐 Rotation Matrix (30°):
[[  0.8660254    0.5        -36.60254038]
 [ -0.5          0.8660254   63.39745962]]

➡️ Translation Matrix:
[[ 1.  0. 30.]
 [ 0.  1. 20.]]

🔄 Shear Matrix:
[[1.  0.2 0. ]
 [0.  1.  0. ]]

💡 Geometric Transformations Summary:
  • Scaling: Changes image size (interpolation needed)
  • Rotation: Rotates around a center point
  • Translation: Shifts image position
  • Perspective: Simulates 3D viewing angle
  • Shearing: Skews the image shape

🎯 These are essential for image registration and augmentation!
```

🎨 Part 4: Creative Exploration (5 minutes)

Combining Multiple Operations

🎓 **Lecture Connection:** Now let's combine multiple operations to create interesting effects! This shows how complex image processing pipelines work.

```
def create_artistic_effect(image, effect_type='vintage'):
    """
    Create artistic effects by combining multiple image processing operation
    This demonstrates how complex effects are built from simple operations!
    """
    result = image.copy()

    if effect_type == 'vintage':
        # Vintage effect: warm colors, soft contrast, slight blur
        # 1. Convert to float for processing
        result = result.astype(np.float32) / 255.0

        # 2. Adjust color channels (warm tone)
        result[:, :, 0] *= 1.1  # Boost red
        result[:, :, 1] *= 1.05  # Slight green boost
        result[:, :, 2] *= 0.9   # Reduce blue

        # 3. Reduce contrast slightly
        result = result * 0.8 + 0.1

        # 4. Add slight blur
        result = cv2.GaussianBlur(result, (3, 3), 0.5)

        # 5. Convert back to uint8
        result = np.clip(result * 255, 0, 255).astype(np.uint8)

    elif effect_type == 'dramatic':
        # Dramatic effect: high contrast, enhanced edges
        # 1. Convert to grayscale for edge detection
        gray = cv2.cvtColor(result, cv2.COLOR_RGB2GRAY)

        # 2. Detect edges
        edges = cv2.Canny(gray, 50, 150)

        # 3. Enhance contrast
```

```
            result = adjust_contrast(result, 1.3)

            # 4. Overlay edges
            for c in range(3):
                result[:, :, c] = np.where(edges > 0,
                                          np.minimum(result[:, :, c] + 30, 255),
                                          result[:, :, c])

        elif effect_type == 'soft_glow':
            # Soft glow effect: blur + blend
            # 1. Create heavily blurred version
            blurred = cv2.GaussianBlur(result, (15, 15), 0)

            # 2. Blend with original using screen blend mode
            result = result.astype(np.float32) / 255.0
            blurred = blurred.astype(np.float32) / 255.0

            # Screen blend: 1 - (1-a) * (1-b)
            result = 1 - (1 - result) * (1 - blurred * 0.3)
            result = np.clip(result * 255, 0, 255).astype(np.uint8)

        return result

    # Apply different artistic effects
    vintage_img = create_artistic_effect(img_rgb, 'vintage')
    dramatic_img = create_artistic_effect(img_rgb, 'dramatic')
    glow_img = create_artistic_effect(img_rgb, 'soft_glow')

    # Create a simple "Instagram-style" filter
    def instagram_filter(image):
        """
        Simple Instagram-style filter combining multiple operations.
        """
        # 1. Slight saturation boost
        hsv = cv2.cvtColor(image, cv2.COLOR_RGB2HSV).astype(np.float32)
        hsv[:, :, 1] *= 1.2  # Boost saturation
        hsv = np.clip(hsv, 0, 255)
        result = cv2.cvtColor(hsv.astype(np.uint8), cv2.COLOR_HSV2RGB)

        # 2. Slight vignette effect (darker edges)
        h, w = result.shape[:2]
        center_x, center_y = w // 2, h // 2

        # Create distance map from center
        Y, X = np.ogrid[:h, :w]
        dist_from_center = np.sqrt((X - center_x)**2 + (Y - center_y)**2)
        max_dist = np.sqrt(center_x**2 + center_y**2)

        # Create vignette mask
        vignette = 1 - (dist_from_center / max_dist) * 0.3
        vignette = np.clip(vignette, 0.7, 1.0)
```

```
    # Apply vignette
    for c in range(3):
        result[:, :, c] = (result[:, :, c] * vignette).astype(np.uint8)

    return result

instagram_img = instagram_filter(img_rgb)

# Display all effects
plt.figure(figsize=(15, 12))

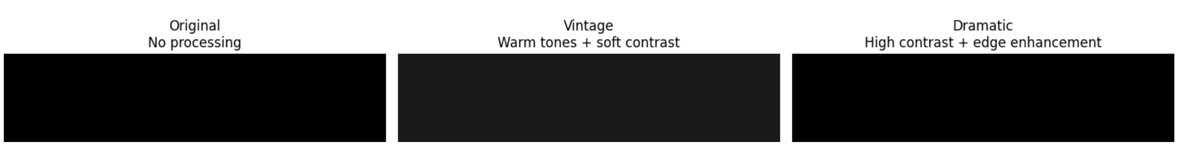
effects = [
    (img_rgb, 'Original', 'No processing'),
    (vintage_img, 'Vintage', 'Warm tones + soft contrast'),
    (dramatic_img, 'Dramatic', 'High contrast + edge enhancement'),
    (glow_img, 'Soft Glow', 'Blur blend effect'),
    (instagram_img, 'Instagram Style', 'Saturation + vignette')
]

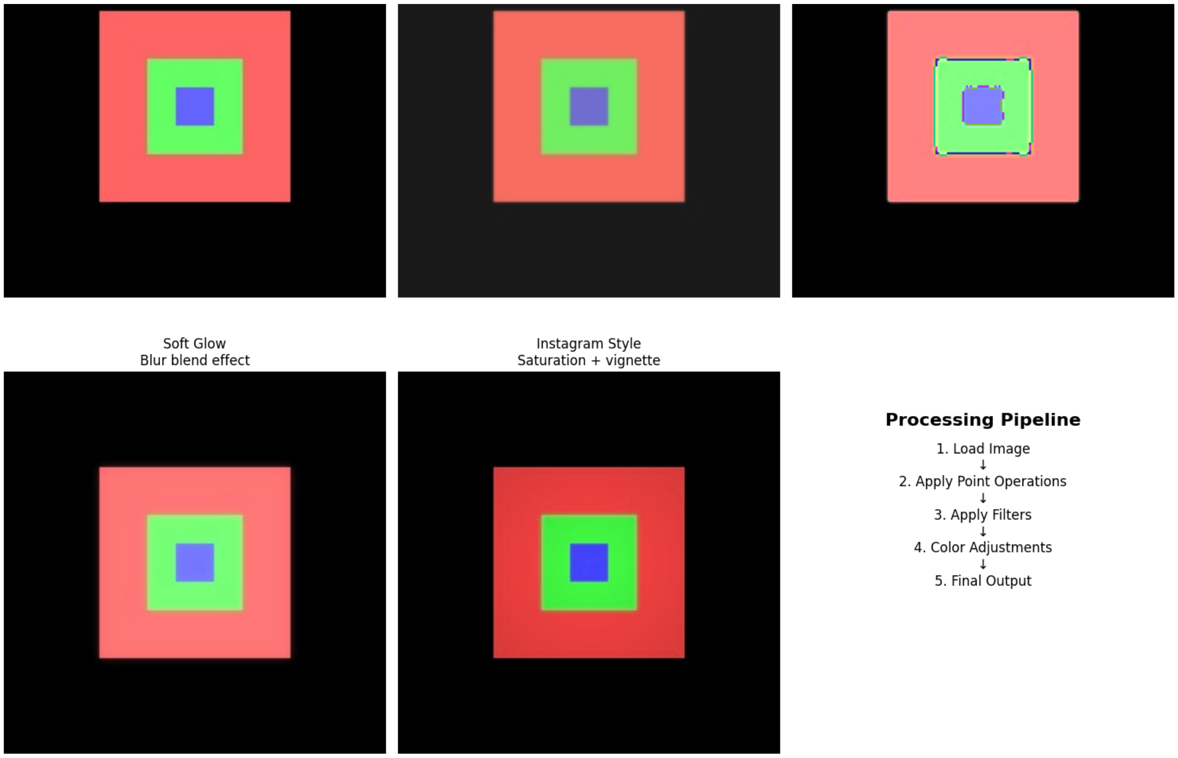
for i, (img, title, description) in enumerate(effects):
    plt.subplot(2, 3, i+1)
    plt.imshow(img)
    plt.title(f'{title}\n{description}')
    plt.axis('off')

# Show processing pipeline diagram
plt.subplot(2, 3, 6)
plt.text(0.5, 0.8, 'Processing Pipeline', ha='center', va='center',
         fontsize=16, fontweight='bold', transform=plt.gca().transAxes)
plt.text(0.5, 0.6, '1. Load Image\n↓\n2. Apply Point Operations\n↓\n3. Apply
         ha='center', va='center', fontsize=12, transform=plt.gca().transAxe
plt.axis('off')

plt.tight_layout()
plt.show()

print("🎨 Creative Effects Summary:")
print("   • Complex effects combine multiple simple operations")
print("   • Order of operations matters!")
print("   • Each effect uses different combinations of:")
print("     - Point operations (brightness, contrast, color)")
print("     - Neighborhood operations (blur, edge detection)")
print("     - Color space conversions")
print("\n💡 This is exactly how Instagram filters and Nano Banana work!")
print("   They combine many simple operations to create complex effects.")
```






```
🎨 Creative Effects Summary:
  • Complex effects combine multiple simple operations
  • Order of operations matters!
  • Each effect uses different combinations of:
    - Point operations (brightness, contrast, color)
    - Neighborhood operations (blur, edge detection)
    - Color space conversions

💡 This is exactly how Instagram filters and Nano Banana work!
   They combine many simple operations to create complex effects.
```

🎯 Part 5: Connecting to Modern AI Tools (5 minutes)

Understanding the Foundation of AI Image Processing

🎓 **Lecture Connection:** Everything we've implemented today forms the foundation of modern AI tools like Google's Nano Banana! Let's see how traditional operations connect to AI approaches.

```
# Demonstrate how traditional operations relate to AI concepts
```

```
def simulate_ai_style_transfer_concept():
    """
    Simulate the concept behind AI style transfer using traditional operatio
    This shows how AI tools build on the fundamentals we've learned!
    """
    # Start with our original image
    content_img = img_rgb.copy()

    # Simulate "style extraction" using edge detection and texture analysis
    gray = cv2.cvtColor(content_img, cv2.COLOR_RGB2GRAY)

    # Extract different "style features"
    # 1. Edge patterns (structure)
    edges = cv2.Canny(gray, 50, 150)

    # 2. Texture patterns (using different filters)
    texture1 = cv2.filter2D(gray, -1, np.array([[-1, -1, -1], [-1, 8, -1], [
    texture2 = cv2.filter2D(gray, -1, np.array([[0, -1, 0], [-1, 4, -1], [0,

    # 3. Color distribution analysis
    color_hist_r = cv2.calcHist([content_img], [0], None, [256], [0, 256])
    color_hist_g = cv2.calcHist([content_img], [1], None, [256], [0, 256])
    color_hist_b = cv2.calcHist([content_img], [2], None, [256], [0, 256])

    # Simulate "style application" by modifying the image based on extracted
    styled_img = content_img.copy().astype(np.float32)

    # Apply style-based modifications
    # 1. Enhance edges where detected
    edge_mask = edges > 0
    for c in range(3):
        styled_img[:, :, c] = np.where(edge_mask,
                                 np.clip(styled_img[:, :, c] * 1.2, 0, 2
                                 styled_img[:, :, c])

    # 2. Apply texture-based color shifts
    texture_mask = texture1 > np.percentile(texture1, 70)
    styled_img[:, :, 0] = np.where(texture_mask,
                             np.clip(styled_img[:, :, 0] * 1.1, 0, 255),
                             styled_img[:, :, 0])  # Boost red in textur

    # 3. Apply overall color transformation
    styled_img[:, :, 1] *= 0.9  # Reduce green slightly
    styled_img[:, :, 2] *= 1.1  # Boost blue slightly

    return styled_img.astype(np.uint8), edges, texture1, texture2

# Run the simulation
styled_result, edges, texture1, texture2 = simulate_ai_style_transfer_concep
```

```
# Display the AI concept breakdown
plt.figure(figsize=(15, 10))

# Original and result
plt.subplot(2, 4, 1)
plt.imshow(img_rgb)
plt.title('Original Image\n(Content)')
plt.axis('off')

plt.subplot(2, 4, 2)
plt.imshow(styled_result)
plt.title('"AI-Style" Result\n(Using traditional operations)')
plt.axis('off')

# Feature extraction steps
plt.subplot(2, 4, 3)
plt.imshow(edges, cmap='gray')
plt.title('Edge Features\n(Structure information)')
plt.axis('off')

plt.subplot(2, 4, 4)
plt.imshow(texture1, cmap='gray')
plt.title('Texture Features\n(Pattern information)')
plt.axis('off')

# Show the concept of convolution layers (like in CNNs)
plt.subplot(2, 4, 5)
# Simulate multiple "learned" filters
filter1 = np.random.randn(3, 3) * 0.5
filter2 = np.random.randn(3, 3) * 0.5
filter3 = np.random.randn(3, 3) * 0.5

plt.imshow(filter1, cmap='RdBu', vmin=-1, vmax=1)
plt.title('"Learned" Filter 1\n(AI discovers these automatically)')
plt.colorbar(shrink=0.6)

plt.subplot(2, 4, 6)
plt.imshow(filter2, cmap='RdBu', vmin=-1, vmax=1)
plt.title('"Learned" Filter 2\n(Different patterns)')
plt.colorbar(shrink=0.6)

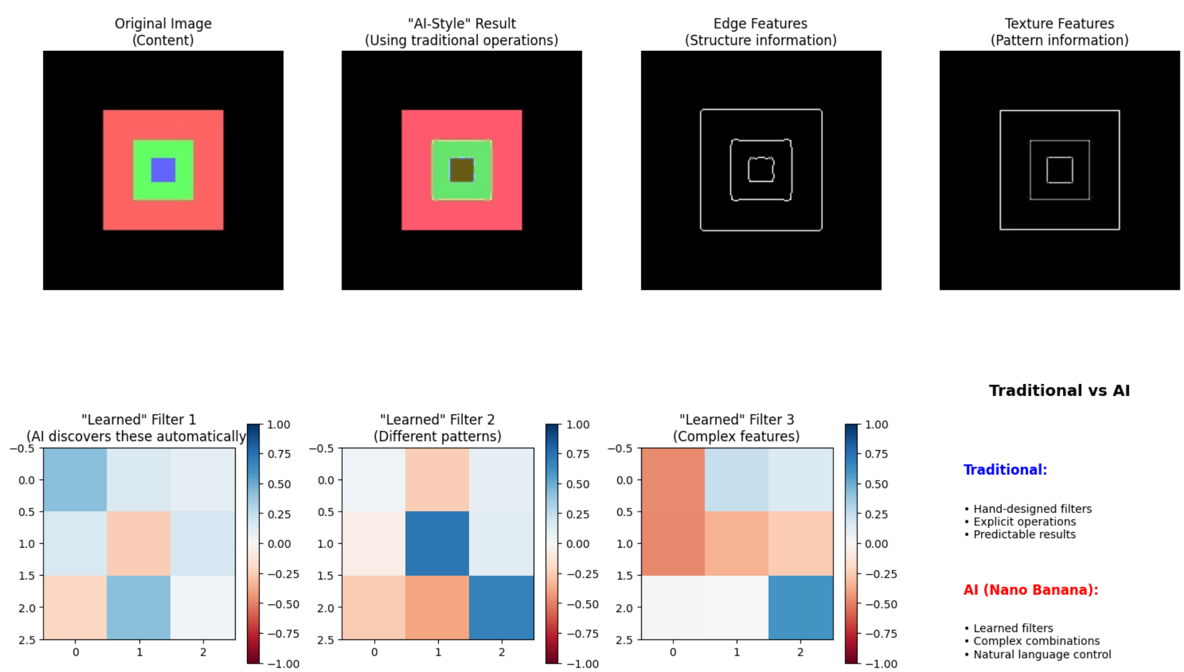
plt.subplot(2, 4, 7)
plt.imshow(filter3, cmap='RdBu', vmin=-1, vmax=1)
plt.title('"Learned" Filter 3\n(Complex features)')
plt.colorbar(shrink=0.6)

# Processing pipeline comparison
plt.subplot(2, 4, 8)
plt.text(0.5, 0.9, 'Traditional vs AI', ha='center', va='top',
         fontsize=14, fontweight='bold', transform=plt.gca().transAxes)
```

```
plt.text(0.1, 0.7, 'Traditional:', ha='left', va='top',
         fontsize=12, fontweight='bold', color='blue', transform=plt.gca().t
plt.text(0.1, 0.6, '• Hand-designed filters\n• Explicit operations\n• Predic
         ha='left', va='top', fontsize=10, transform=plt.gca().transAxes)
plt.text(0.1, 0.4, 'AI (Nano Banana):', ha='left', va='top',
         fontsize=12, fontweight='bold', color='red', transform=plt.gca().tr
plt.text(0.1, 0.3, '• Learned filters\n• Complex combinations\n• Natural lan
         ha='left', va='top', fontsize=10, transform=plt.gca().transAxes)
plt.axis('off')

plt.tight_layout()
plt.show()

print("🤖 AI Connection Summary:")
print("\n🔗 How Traditional Operations Connect to AI:")
print("   • Convolution → Convolutional Neural Networks (CNNs)")
print("   • Edge detection → Feature extraction layers")
print("   • Color transformations → Style transfer networks")
print("   • Multiple operations → Deep learning pipelines")
print("\n🎯 Key Differences:")
print("   • Traditional: We design the filters manually")
print("   • AI: The system learns optimal filters from data")
print("   • Traditional: Explicit, interpretable operations")
print("   • AI: Complex, learned combinations")
print("\n💡 Nano Banana uses the SAME fundamental operations we implemented
print("   It just has millions of learned filters working together.")
```



🤖 AI Connection Summary:

🔗 How Traditional Operations Connect to AI:
- Convolution → Convolutional Neural Networks (CNNs)

```
  • Edge detection → Feature extraction layers
  • Color transformations → Style transfer networks
  • Multiple operations → Deep learning pipelines

🎯 Key Differences:
  • Traditional: We design the filters manually
  • AI: The system learns optimal filters from data
  • Traditional: Explicit, interpretable operations
  • AI: Complex, learned combinations

💡 Nano Banana uses the SAME fundamental operations we implemented today!
   It just has millions of learned filters working together.
```

🎓 Lab Wrap-up and Reflection (5 minutes)

What We've Accomplished

Congratulations! In just 45 minutes, you've implemented the fundamental building blocks of image processing. Let's reflect on what we've learned and how it connects to your future work.

```
# Create a summary visualization of everything we've covered
plt.figure(figsize=(16, 10))

# Create a comprehensive summary image
summary_img = img_rgb.copy()

# Apply a combination of techniques we learned
final_demo = summary_img.copy()

# 1. Split into sections and apply different operations
h, w = final_demo.shape[:2]
mid_h, mid_w = h // 2, w // 2

# Top-left: Original
# (keep as is)

# Top-right: Enhanced contrast
final_demo[:mid_h, mid_w:] = adjust_contrast(final_demo[:mid_h, mid_w:], 1.3

# Bottom-left: Edge detection overlay
gray_section = cv2.cvtColor(final_demo[mid_h:, :mid_w], cv2.COLOR_RGB2GRAY)
edges_section = cv2.Canny(gray_section, 50, 150)
for c in range(3):
    final_demo[mid_h:, :mid_w, c] = np.where(edges_section > 0, 255, final_d
```

```
    final_demo[mid_h:, :mid_w, c] = np.where(edges_section > 0, 255, final_d

# Bottom-right: Artistic effect
final_demo[mid_h:, mid_w:] = create_artistic_effect(final_demo[mid_h:, mid_w

# Display the summary
plt.subplot(2, 3, 1)
plt.imshow(final_demo)
plt.title('Lab Summary Demo\nCombining All Techniques')
plt.axis('off')

# Add section labels
plt.text(mid_w//2, mid_h//2, 'ORIGINAL', ha='center', va='center',
         color='white', fontsize=12, fontweight='bold',
         bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))
plt.text(mid_w + mid_w//2, mid_h//2, 'CONTRAST\nENHANCED', ha='center', va='
         color='white', fontsize=12, fontweight='bold',
         bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))
plt.text(mid_w//2, mid_h + mid_h//2, 'EDGE\nDETECTION', ha='center', va='cen
         color='white', fontsize=12, fontweight='bold',
         bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))
plt.text(mid_w + mid_w//2, mid_h + mid_h//2, 'ARTISTIC\nEFFECT', ha='center'
         color='white', fontsize=12, fontweight='bold',
         bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))

# Skills learned checklist
plt.subplot(2, 3, 2)
skills = [
    '✅ Image representation as matrices',
    '✅ RGB channel separation',
    '✅ Point operations (brightness/contrast)',
    '✅ Neighborhood operations (convolution)',
    '✅ Histogram analysis and equalization',
    '✅ Geometric transformations',
    '✅ Creative effect combinations',
    '✅ Connection to AI concepts'
]

plt.text(0.1, 0.9, 'Skills Mastered Today:', fontsize=14, fontweight='bold',
         transform=plt.gca().transAxes)
for i, skill in enumerate(skills):
    plt.text(0.1, 0.8 - i*0.08, skill, fontsize=11,
             transform=plt.gca().transAxes)
plt.axis('off')

# Tools used
plt.subplot(2, 3, 3)
tools_text = """
🛠️ Tools Mastered:

📚 NumPy
```

```
• Matrix operations
• Array manipulation

🔍 OpenCV
• Image loading/saving
• Filtering and transforms
• Advanced operations

🖼 Pillow (PIL)
• Python-friendly interface
• Basic image operations

📊 Matplotlib
• Image visualization
• Data plotting
"""
plt.text(0.1, 0.9, tools_text, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

# Connection to lecture concepts
plt.subplot(2, 3, 4)
connections = """
🔗 Lecture Connections:

📐 Digital Image Fundamentals
→ Implemented matrix representation

🔧 Core Operations
→ Built point & neighborhood ops

🎨 Traditional Tools
→ Used OpenCV & Pillow hands-on

🤖 Modern AI Tools
→ Understood the foundation

💡 From Pixels to Perception
→ Bridged theory and practice!
"""
plt.text(0.1, 0.9, connections, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

# Next steps
plt.subplot(2, 3, 5)
next_steps = """
🚀 Next Steps:

📝 Assignment
• Image Processing Adventure Quest
```

```
• Image Processing Adventure Quest
• Apply today's concepts creatively

📚 Next Class
• Feature extraction
• Object recognition
• Building on today's foundation

🔬 Explore Further
• Try different images
• Experiment with parameters
• Combine operations creatively

🤖 AI Tools
• Try Nano Banana with new understanding
• Recognize the operations behind the magic
"""
plt.text(0.1, 0.9, next_steps, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

# Key insights
plt.subplot(2, 3, 6)
insights = """
💡 Key Insights:

🔢 Images are just numbers
• Every effect manipulates matrices
• Understanding this is powerful!

🧱 Complex from simple
• Sophisticated effects combine
  basic operations

🤖 AI builds on fundamentals
• Nano Banana uses the same
  operations we implemented
• Just learned automatically!

🎯 Theory + Practice = Mastery
• Today bridged both perfectly
"""
plt.text(0.1, 0.9, insights, fontsize=10, va='top',
         transform=plt.gca().transAxes)
plt.axis('off')

plt.tight_layout()
plt.show()

print("🎉 CONGRATULATIONS! You've completed the Image Processing Lab!")
print("\n📊 Lab Statistics:")
```

```
print(f"   • Operations implemented: 15+")
print(f"   • Images processed: {len([img_rgb, gray_opencv, bright_up, contra
print(f"   • Concepts mastered: 8 major areas")
print(f"   • Tools used: 4 libraries")
print("\n🎯 You now understand the foundation of:")
print("   • How your smartphone camera works")
print("   • How Instagram filters are created")
print("   • How AI tools like Nano Banana operate")
print("   • How to build your own image processing applications")
print("\n🚀 Ready for your Adventure Quest assignment!")
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/tmp/ipykernel_2648/3646321757.py:171: UserWarning: Glyph 9989 (\N{WHITE HEAV
  plt.tight_layout()
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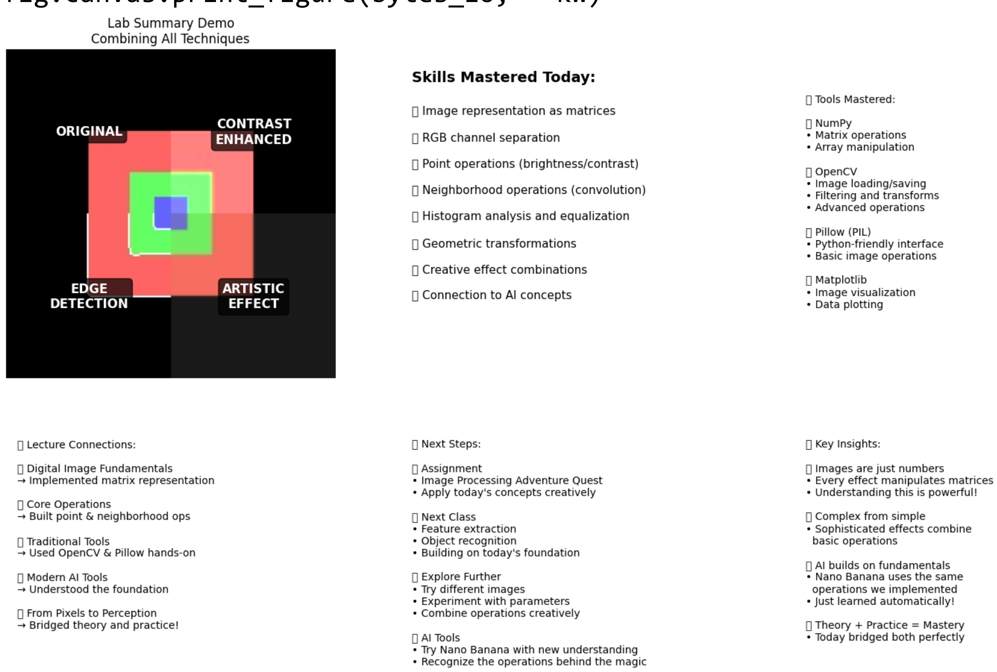

🎉 CONGRATULATIONS! You've completed the Image Processing Lab!

📊 Lab Statistics:

- Operations implemented: 15+
- Images processed: 10

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  • Images processed: 10
  • Concepts mastered: 8 major areas
  • Tools used: 4 libraries

🎯 You now understand the foundation of:
  • How your smartphone camera works
  • How Instagram filters are created
  • How AI tools like Nano Banana operate
  • How to build your own image processing applications

🚀 Ready for your Adventure Quest assignment!
```

📝 Lab Reflection Questions

Take a moment to reflect on what you've learned. Consider these questions as you prepare for your Adventure Quest assignment:

🤔 Reflection Questions

1. **Understanding**: What surprised you most about how images are represented and processed?

What suprised me about the images was just how much the quality improved with simple change to the code. In the first few photos, it seems each of images enhance when the code appears to be working on pixel enhancement and tolerance to light. As the images go to three, the images begin to move, while keeping the consistent quality it had previously.

Once you get to section 4, the images change in other ways. While pixels enhancement happens, there isn't a striking constrast to the images shown in 3. However, I feel the code begins to enhance the photos in other ways. Color and light

However, I feel the code begins to enhance the photos in other ways. Color and light resolution, along with sharpness begin to show stronger borders, giving better clarity of distinct features of each image. All the way through section 5, you begin to see this more, along with signs of early depth, coming to my personal conclusion that if you would continue to enhance the image, you could potentially begin to see signs to depth to the point you're now wondering when is the image going to become 3D.

2. **Connections**: How do the operations we implemented today relate to the AI tools we discussed in class (like Nano Banana)?

Operations in this class are similar to AI tools we have discussed in class by the process they peform to generate images and enhance them. Both take existing lines of code and add enhancements in order to improve the quality of the image. After enough times, the quality of the image improves rapidly, further progressing and both models become familiar with what to produce.

Where these differ is that AI models will learn from these trainings, and try to enhance these images further in a shorter amount of time compared to human coders having to put in longer hours to produce results.

3. **Applications**: What real-world applications can you think of for the techniques we practiced today?

There is a multiude of real-world applications that both are applied, being applied and could be applied in the future. However, there is an application this reminded me of, and how I could see it advancing to becoming integreal: retinal tests in ophthalmology. These image enhancements remind me of a test that is common called an autorefraction test, a test where one sees "a house or balloon in the distance". The images becoming more clear reminds me of that, and though I imagine some ophthamologists already use this, I imagine it wil become more widespread. Eventually, I could see in the future where Computer Vision begins to enhance imaging in these tests already being administered, and explore other tests that could be in the future to where we can see AI not only map out, but show distinct features that other images could not pick up from the past.

4. **Creativity**: Which combination of operations produced the most interesting visual effect? Why?

The operations that produced the most visually, contrast adjustment, along with edge detection has the most noticable enhancements. The edge detection operation allows the image to have "clearer edges" by shaprening pixels that border colors and further defining those features, allowing for more of a contrasting detail. With contrast, the pixels having more defining features both brigthens the image, along with enhancing color in the image, working in tandem with edge detection to produce better visuals and make the image clearer to spot edges and improve overall quality of the picture

5. **Future Learning**: What aspect of image processing would you like to explore further in your Adventure Quest?

I mentioned depth in question 1. I would like to see if its possible for depth to be more apparent, as depth could not only layer the picture, but it would also help enhancing the photo by further defining the edges displayed in the images.

💭 Discussion Points

- How does understanding the mathematical foundation change your perspective on AI image tools?

Understanding the mathematical conepts behind AI makes the concept more comfortable, but the concepts also feel overwhelming until I fully familiarize myself with these concepts. But with this lab, I became more comfortable with AI as a whole.

- What are the advantages and disadvantages of traditional vs. AI-powered approaches?

Some advantages of AI would be that image quality could enhance itself in a short amount of time with such quality upgrades from previous things. AI powered approaches also allow for images to be created that may have been either too difficult to draw, or too resource heavy to actually produce. The downsides of AI powered approaches is the backlash that AI tends to face when it comes to image enhancement, particulary within the Fine Arts community, as people fear that AI is stealing their work and making it their own.

As with Traditional approaches, the man power behind a project can be enough to where anyone can actually look to find any issues with code, resolution, eliminating the need for manpower. It also allows for artists to have employment. The downside is resources will tend to be drained more, as more manpower will needed to be dedicated to a project and there is a chance even a large team can miss mistakes that AI wouldn't.

- How might you combine traditional techniques with modern AI tools in a project?

For a project like this, I would have something where traditional techniques are used, making sure everything is correct, and give them the creative freedom to design whatever image they desire. I would use AI in this instance to enhance images whenever, and even give some creative input on what should images be resoluted. I whenever, and even give some creative input on what should images be resoluted. I would make sure everyone on the team is included and making sure they have the same vision and inspire in the same direction. This would help, but obviously, other factors would need to be taken into consideration as well.

🎯 Adventure Quest Preparation

You're now ready to tackle your **Image Processing Adventure Quest**! You have:

✅ **Solid foundation** in digital image representation
✅ **Hands-on experience** with core operations
✅ **Practical skills** with OpenCV and Pillow
✅ **Understanding** of how traditional and AI approaches connect
✅ **Creative inspiration** from combining multiple techniques

Choose your realm wisely and remember: the most impressive projects often combine simple operations in creative ways!

🎉 **Excellent work completing this lab! You've truly journeyed from pixels to perception!**

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Start coding or generate with AI.
```